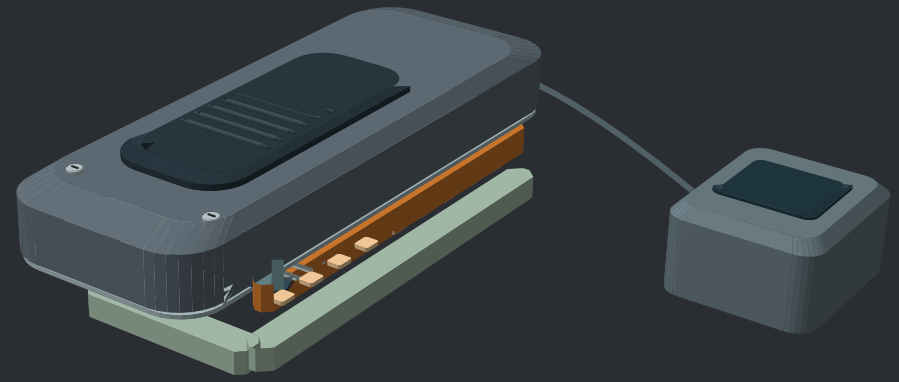
# VASCOUT

## See anatomy. Understand uncertainty.

A proposed ultrasound-led vascular imaging system for clinician-controlled assessment in difficult-access and resource-limited settings.

**DESIGN TARGET / DEVICE UNVALIDATED**

**Explore the live concept and evidence**



ILLUSTRATIVE PROBE + SEPARATE ACQUISITION UNIT

# Difficult access deserves clearer evidence.

Research intent: test whether anatomy and image-quality feedback can support clinician assessment.

CLINICAL CONTEXT / S10a + S10b

## Difficult peripheral access

Published ultrasound-access findings differ by workflow and operator. The comparator for VASCOUT should be ultrasound alone, with a defined task and ground truth.

FIELD RESEARCH / UNVERIFIED

## Low perfusion. Limited resources.

Dehydration, shock, motion, gloves and limited resources motivate proposed studies. VASCOUT benefit under these conditions has not been measured.

## The intended contribution

Make anatomy and uncertainty visible for review. The concept does not select an insertion site, prescribe treatment or demonstrate successful IV access.

S10a / SOURCE  S10b / SOURCE  S07 / SOURCE

# Ultrasound first. Added contrast to test.

Visible transillumination, NIR camera imaging and photoacoustic tomography are distinct methods. [S01]

01 / ACQUIRE

## Ultrasound structure

Candidate linear array; coupling and low contact force.

>

02 / COMPARE

## Experimental PA contrast

Test US alone against US + PA under matched conditions.

>

03 / REVIEW

## Anatomy + uncertainty

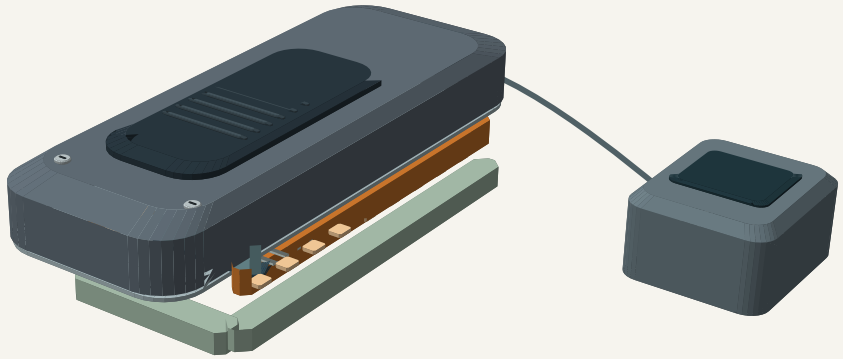
Clinician interprets anatomy and invalid or missing signal.

TARGET: synchronized acquisition, registration and explicit quality feedback. Added value, latency, exposure and thermal budgets remain UNVERIFIED.

S01 / SOURCE S02 / SOURCE S03 / SOURCE

# A probe within a larger research system.

Illustrative geometry from the interactive VASCOUT model. No built prototype or measured packaging performance.



01 / HANDLE

## Deliberate contact

Grip, controls and proposed force feedback need usability studies.

02 / INTERFACE

## Passive contact shoe

Open corridor; acoustic and optical transmission require measurement.

03 / SYSTEM

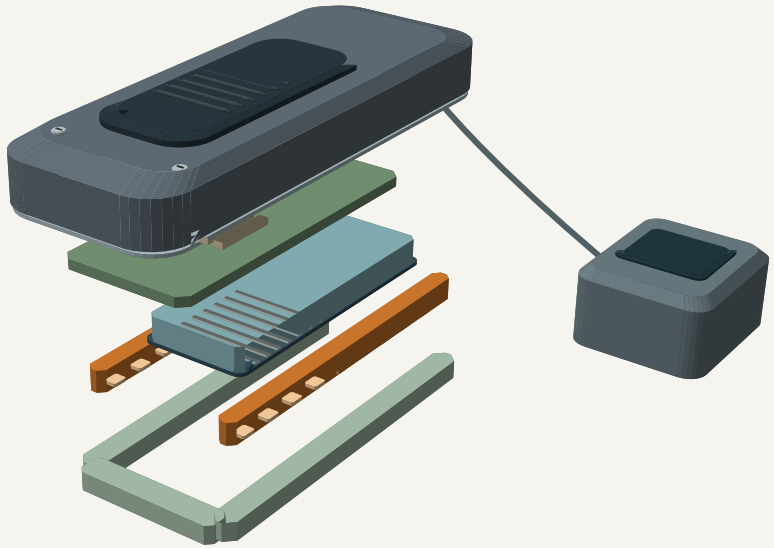
## Separate acquisition unit

The source, acquisition chain and power budget are not contained in the handle.

TARGET / ILLUSTRATIVE / NOT TO SCALE

# Five interfaces to engineer and test.

Layer separation is illustrative; materials, interfaces and integration remain design work.



01 **Housing**

Gloved reach, cleaning, balance and contact-force usability studies.

02 **Acoustic sensor**

Point-spread function, vessel localization, near-field performance and deformation.

03 **Optical rails**

Incremental benefit over US alone, exposure, heating, pigmentation effects and motion.

04 **Contact shoe**

Transmission loss, air gaps, vessel compression, cleaning and sterility requirements.

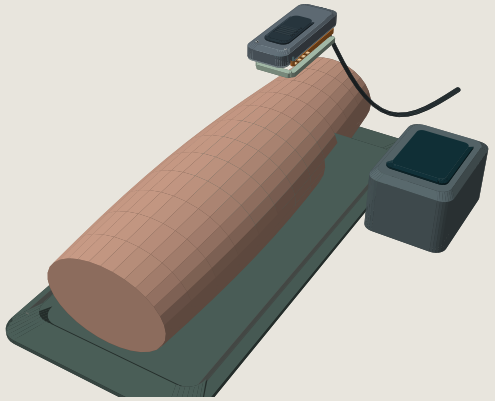
05 **Electronics**

Synchronization, registration, failure states, latency, power and thermal budgets.

SEPARATE SOURCE / ACQUISITION UNIT REQUIRED

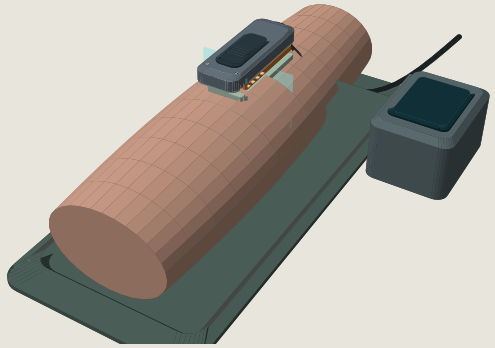
# From a sweep to clinician review.

Static frames from a proposed 30-second workflow. Synthetic anatomy, pre-authored vessels and no measured patient data.



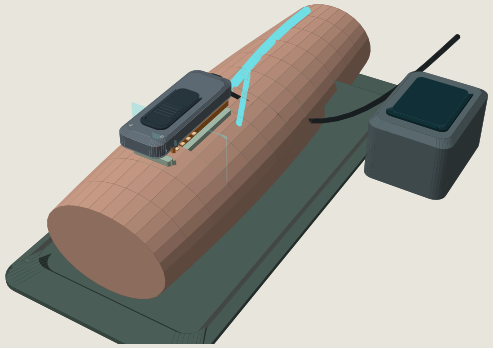
00s / FIELD SETUP

Proposed probe, tether and field pad.



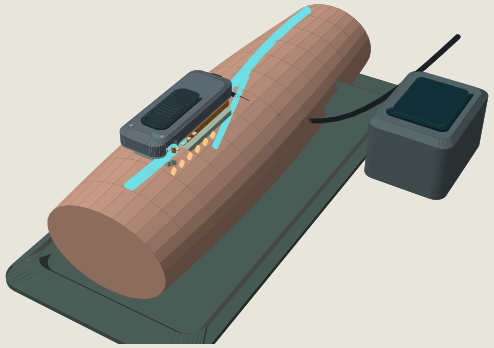
10s / SIMULATED SCAN

Drawn scan plane; no ultrasound acquisition.



19s / VESSEL MAP

Pre-authored path; no vessel detected.



23s / CLINICIAN REVIEW

Unresolved anatomy; no IV outcome.

CONCEPT ONLY: no scan generated, no vessel measured, no insertion guidance and no demonstrated access success.

**Open the live animated concept**

# Numbers are inputs to investigation.

All values below are TARGETS or experiment inputs. They are not demonstrated device specifications.

| | | |
|---|---|---|
| **Probe envelope** | 150 x 56 x 34 mm | Illustrative packaging; ergonomics and integration pending. |
| **Ultrasound band** | 7-15 MHz candidate | Hardware selection and measured imaging response pending. |
| **PA illumination** | 850 nm LED / 1064 nm pulsed | Alternative laboratory configurations; no combined claim. |
| **Phantom depths** | 2 / 5 / 10 / 15 mm | Test positions; not validated operating depths. |
| **Phantom diameters** | 1 / 2 / 4 / 6 mm | Test sizes; not clinical access thresholds. |
| **3D acquisition** | Tracked sweep / bench scan | Registration and motion sensitivity unmeasured. |
| **Contact / invalid data** | Passive shoe + force / UNKNOWN | Deformation, coupling, safety and failure modes to test. |

**NO DEPTH, FRAME-RATE, SAFETY OR FIELD-READY PERFORMANCE CLAIM**

# What VASCOUT must still establish.

Current position: concept design and literature review. Every device-level gate below is UNVERIFIED.

01 / IMAGING / UNVERIFIED

## Resolution, localization and coverage

Benchmark against known geometry across the phantom matrix.

02 / HYBRID BENEFIT / UNVERIFIED

## Incremental benefit over ultrasound

Define a task, comparator, success criteria and added time.

03 / EQUITY + QUALITY / UNVERIFIED

## Pigmentation, motion and signal dropout

Measure failure modes; no skin-tone invariance established. [S06]

04 / CONTACT + SAFETY / UNVERIFIED

## Deformation, exposure and heating

Measure optical/acoustic exposure, force and thermal budgets.

05 / FIELD USABILITY / UNVERIFIED

## Gloves, cleaning and constrained use

Simulated use before any assertion of field suitability.

06 / CLINICAL + REGULATORY / UNVERIFIED

## Intended use and evidence pathway

Expert review and appropriate approvals before human studies. [S11]

S06 / SOURCE

S11 / SOURCE

# A path from concept to evidence.

Advancement depends on measured results and predefined failure criteria; no stage is marked complete.

| 01 | 02 | 03 | 04 | 05 |
|---|---|---|---|---|
| **BENCH COMPARISON** | **CALIBRATED PHANTOMS** | **SIMULATED USE** | **EXPERT REVIEW** | **APPROVED HUMAN STUDIES** |
| Choose components, ground truth, calibration and US-only comparator. | Measure localization, diameter error, dropout, bias and deformation. | Test gloves, motion, lighting, cleaning, setup and failure recognition. | Refine intended use, risk controls and study requirements. | Only after appropriate approvals; evaluate safety and intended benefit. |
| UNVERIFIED | UNVERIFIED | UNVERIFIED | UNVERIFIED | UNVERIFIED |

Decision at each gate: proceed, revise or stop based on agreed evidence. Foreign-body localization remains a separate future phantom study; negative imaging cannot exclude fragments. [S08, S09]

# Define the first study together.

A research collaboration discussion for medical companies, imaging developers and defense medical organizations.

IMAGING + HARDWARE

## Build the benchmark

Array/source selection, calibrated phantoms, acquisition and ground truth.

MEDICAL + CLINICAL

## Define the intended task

Clinician workflow, useful outputs, comparator and meaningful failure criteria.

DEFENSE MEDICAL

## Shape simulated conditions

Gloves, movement, resource constraints, setup, cleaning and usability requirements.

---

## First working session

Agree the comparator, ground truth, failure criteria and evidence required to proceed.

**Explore the device and evidence library**

# Eight prospects. A defined first conversation.

Qualitative fit assessment / INFERENCE. No affiliation, endorsement, partner interest or funding confirmed.

| | | | |
|---|---|---|---|
| 01 | **Verasonics** | Programmable US/PA bench platform | P01 / P02 |
| 02 | **CYBERDYNE / Acoustic X** | LED-PA comparator and prior art | P03 / P04 |
| 03 | **Michigan / Optical Imaging Lab** | Independent imaging methodology | P14 |
| 04 | **U.S. Army Institute of Surgical Research** | Military-medical requirements and simulation | P11 / P12 |
| 05 | **FUJIFILM Sonosite** | POCUS workflow and image-quality research | P05 / P06 |
| 06 | **BD** | Vascular-access comparator and workflow | P07 / P08 |
| 07 | **Teleflex / Arrow** | Access workflow and contact-interface usability | P09 / P10 |
| 08 | **Medical Technology Enterprise Consortium** | Teaming and funding-route assessment | P13 |

Suggested first discussions: Verasonics, CYBERDYNE and Michigan for a bench comparator and methods critique. This sequence is an inference, not an acceptance or funding ranking.

ALL PITCHES ARE DRAFTS / PREPARED WITH AI / NOT SENT

# Verasonics

Programmable US/PA bench platform / Proposed research fit is an INFERENCE. Partner interest is unconfirmed.

### SOURCED CAPABILITY / P01, P02

Verasonics describes research ultrasound with receive-only photoacoustic operation and interleaved B-mode/PAI. It explicitly invites collaborations with companies and researchers.

### PROPOSED FIRST STUDY / INFERENCE

Scope an ultrasound-only versus US+PA phantom comparison using known vessel geometry, controlled contact force and motion. Measure localization error, dropout, added acquisition time and registration error.

### PROPOSED OUTPUT

A bench architecture, component/trigger compatibility matrix and agreed experimental protocol with stop criteria.

### VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

Hardware compatibility, system configuration, access cost, licensing, staff time and collaboration interest are unverified. A programmable platform is not a deployable field device.

### PUBLISHED ROUTE / NO OUTREACH SENT

Published collaboration/contact route; ask for technical feasibility discussion.

P01 / OFFICIAL SOURCE  P02 / OFFICIAL SOURCE

### TAILORED DRAFT PITCH / PREPARED WITH AI / NOT SENT

VASCOUT is an unvalidated vascular-imaging concept centered on low-contact-force ultrasound, with an experimental photoacoustic extension. Your programmable research platform appears relevant to testing the acquisition chain before custom hardware. Would your team consider a technical scoping discussion for a matched US-only versus US+PA phantom study? We propose agreeing ground truth, trigger compatibility, localization and registration metrics, dropout reporting and a failure threshold before selecting a configuration. The first output would be a reproducible protocol and feasibility matrix; clinical or field benefit is not established.

**Official engagement route**

CLINICAL BENEFIT / SAFETY / FIELD SUITABILITY: UNVERIFIED

# CYBERDYNE / Acoustic X

LED-PA comparator and prior art / Proposed research fit is an INFERENCE. Partner interest is unconfirmed.

### SOURCED CAPABILITY / P03, P04

CYBERDYNE presents Acoustic X as an LED-based PA/US research system. Its page already lists vascular-access-related work and provides an Acoustic X inquiry category.

### PROPOSED FIRST STUDY / INFERENCE

Compare a documented Acoustic X research configuration with ultrasound alone on matched phantoms; vary depth, diameter, optical attenuation, contact force and motion. Treat attenuation phantoms as limited models, not proof across human skin tones.

### PROPOSED OUTPUT

A comparator protocol and an incremental-benefit report that includes negative results, exposure/thermal measurements and acquisition overhead.

### VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

Equipment availability, configuration, data access, research support and willingness to collaborate are unverified. Related prior work means novelty cannot be assumed; no field-access outcome follows.

### PUBLISHED ROUTE / NO OUTREACH SENT

Official product inquiry; select Acoustic X and request research-team routing.

P03 / OFFICIAL SOURCE

P04 / OFFICIAL SOURCE

### TAILORED DRAFT PITCH / PREPARED WITH AI / NOT SENT

Your Acoustic X work already addresses LED-based PA/US and peripheral vascular targets, making it a relevant starting point for VASCOUT rather than evidence that the hybrid idea is new. Would your research team explore a comparator study focused on contact deformation, optical attenuation, motion and explicit invalid-signal reporting? We would agree a US-only baseline and measure incremental localization value, exposure, heating and added time. VASCOUT currently consists of a concept dossier and synthetic demonstration; the proposed collaboration would determine whether its interface and uncertainty-feedback ideas merit further development.

Official engagement route

CLINICAL BENEFIT / SAFETY / FIELD SUITABILITY: UNVERIFIED

# University of Michigan / Optical Imaging Laboratory

Independent imaging methodology / Proposed research fit is an INFERENCE. Partner interest is unconfirmed.

### SOURCED CAPABILITY / P14

The official Michigan BME profile identifies the Optical Imaging Laboratory and research on photoacoustic imaging and systems combining light and ultrasound.

### PROPOSED FIRST STUDY / INFERENCE

Co-design calibrated phantom experiments with independent geometry measurements, optical/acoustic characterization and a prespecified analysis plan. Audit whether image-quality feedback identifies invalid data without overstating anatomical confidence.

### PROPOSED OUTPUT

A reviewed protocol, annotated failure taxonomy and transparent analysis plan; a possible publication would depend on agreement and results.

### VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

Lab capacity, resources, data access and willingness are unconfirmed. The listed research is adjacent expertise, not proof of vascular-access or battlefield performance.

### PUBLISHED ROUTE / NO OUTREACH SENT

Official faculty/lab profile; request an investigator-led methods discussion.

**P14 / OFFICIAL SOURCE**

### TAILORED DRAFT PITCH / PREPARED WITH AI / NOT SENT

VASCOUT proposes clinician-controlled vascular visualization using ultrasound with an experimental PA channel. Your Optical Imaging Laboratory's light-and-ultrasound research appears relevant to a methods review before any clinical claims. Would the group consider helping define phantom ground truth, attenuation and contact-force controls, registration metrics and a prespecified failure analysis? We would value a study designed to find where the added channel fails as well as where it helps. The immediate ask is protocol critique and possible research scoping, with resources and publication terms to be agreed.

**Official engagement route**

CLINICAL BENEFIT / SAFETY / FIELD SUITABILITY: UNVERIFIED

# U.S. Army Institute of Surgical Research

Military-medical requirements and simulation / Proposed research fit is an INFERENCE. Partner interest is unconfirmed.

### SOURCED CAPABILITY / P11, P12

USAISR's CRT3 page lists medical-device development and medical simulations. Its partnering page describes external collaboration and Business Development Office support for CRADAs and technology transfer.

### PROPOSED FIRST STUDY / INFERENCE

After a credible bench protocol, define simulated tasks involving gloves, movement, setup, cleaning and degraded signal. Measure workflow burden and recognition of uncertainty; do not infer battlefield effectiveness from simulation.

### PROPOSED OUTPUT

A research requirements map, simulation protocol and agreed go/no-go criteria tied to measured bench evidence.

### VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

Mission alignment, agreement eligibility, investigator availability, funding and access to facilities are unverified. The public partnering page is not an award or invitation for this device.

### PUBLISHED ROUTE / NO OUTREACH SENT

Research Directorate Business Development Office through the published partnering resources.

P11 / OFFICIAL SOURCE

P12 / OFFICIAL SOURCE

### TAILORED DRAFT PITCH / PREPARED WITH AI / NOT SENT

VASCOUT is a concept-stage, clinician-controlled vascular-imaging proposal; it has no demonstrated field or patient benefit. USAISR's device-development and simulation capabilities appear relevant to defining what evidence would make the concept worth testing. Would the Research Directorate consider a requirements discussion, routed through the appropriate partnering mechanism, on setup burden, contact deformation, degraded-signal recognition and gloved use? We would bring the concept dossier and a proposed US-only comparative bench protocol. The requested output is a measurable simulation plan and explicit reasons to proceed, revise or stop.

**Official engagement route**

CLINICAL BENEFIT / SAFETY / FIELD SUITABILITY: UNVERIFIED

# FUJIFILM Sonosite

POCUS workflow and image-quality research / Proposed research fit is an INFERENCE. Partner interest is unconfirmed.

### SOURCED CAPABILITY / P05, P06

Sonosite's official indexed pages describe military POCUS work and invite clinician/researcher collaboration on POCUS AI, validation, user testing and workflows under research agreements.

### PROPOSED FIRST STUDY / INFERENCE

With a clinical investigator, define a simulated vascular-assessment task comparing ordinary ultrasound review with explicit image-quality/uncertainty feedback. Treat any future AI component as a separately validated research module.

### PROPOSED OUTPUT

A workflow endpoint set, annotation/quality rubric and investigator-led protocol suitable for the program's scope.

### VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

Direct page fetches returned redirect loops; indexed official text supports the lead. Current intake availability, PI eligibility, equipment/support and PA fit need confirmation. No support or funding is promised.

### PUBLISHED ROUTE / NO OUTREACH SENT

Published research-collaboration page; first confirm current program scope and clinician eligibility.

P05 / OFFICIAL SOURCE

P06 / OFFICIAL SOURCE

### TAILORED DRAFT PITCH / PREPARED WITH AI / NOT SENT

VASCOUT is an unvalidated concept for making peripheral anatomy and imaging uncertainty clearer to clinicians. Your published POCUS collaboration route suggests a possible fit for a clinician-led quality-feedback study, rather than an assumption of PA hardware support. Would your research team discuss eligibility and scope for a simulated vascular-assessment protocol comparing ultrasound review with explicit poor-signal feedback? A clinical investigator would define the task and endpoints; any AI component would require separate validation. We seek protocol and workflow feedback first, with resources subject to a formal agreement.

**Official engagement route**

CLINICAL BENEFIT / SAFETY / FIELD SUITABILITY: UNVERIFIED

# BD

Vascular-access comparator and workflow / Proposed research fit is an INFERENCE. Partner interest is unconfirmed.

### SOURCED CAPABILITY / P07, P08

BD describes SiteRite 9 as a vascular-access ultrasound platform and publishes an external Idea Submission Process with a nonconfidential initial review route.

### PROPOSED FIRST STUDY / INFERENCE

Develop a simulation comparison against an agreed vascular-access ultrasound workflow. Assess anatomy localization, contact deformation, interpretation errors, setup burden and invalid-image recognition before any access-success claim.

### PROPOSED OUTPUT

An intended-use/task statement and a comparator protocol with meaningful failure criteria and clinician-defined endpoints.

### VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

Product/integration fit, sponsorship and interest are unverified. BD's initial idea route does not promise confidentiality; detailed submission terms would need review. Existing product capabilities do not validate VASCOUT.

### PUBLISHED ROUTE / NO OUTREACH SENT

Official innovation/idea page; limit any initial discussion to the approved nonconfidential concept.

**P07 / OFFICIAL SOURCE**

**P08 / OFFICIAL SOURCE**

### TAILORED DRAFT PITCH / PREPARED WITH AI / NOT SENT

VASCOUT is a concept-stage ultrasound-led interface with experimental PA contrast and explicit uncertainty feedback. BD's vascular-access portfolio makes the existing clinical workflow an important comparator. Would the relevant research or innovation team consider a nonconfidential scoping discussion on which visualization limitations are worth measuring? We propose a simulated assessment study using predefined localization, deformation, interpretation-error and workflow endpoints. We are not presenting IV-success results or an integrated BD product. The first goal is an agreed task and benchmark protocol, with any detailed exchange governed by appropriate terms.

**Official engagement route**

CLINICAL BENEFIT / SAFETY / FIELD SUITABILITY: UNVERIFIED

# Teleflex / Arrow

Access workflow and contact-interface usability / Proposed research fit is an INFERENCE. Partner interest is unconfirmed.

### SOURCED CAPABILITY / P09, P10

Teleflex publishes Arrow emergency/trauma vascular-access products and an official product inquiry form covering peripheral access and emergency/trauma categories.

### PROPOSED FIRST STUDY / INFERENCE

Use vascular-access task trainers to compare probe/contact-shoe handling, gloved reach, working-corridor obstruction, setup time and cleaning constraints. No actual patient cannulation or IV/IO treatment choice is proposed at this stage.

### PROPOSED OUTPUT

A human-factors requirements sheet and a simulation protocol for contact-interface benefits and failure cases.

### VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

The observed route is a product inquiry, not an announced VASCOUT research program. Research-team routing, sponsorship, compatibility and acceptance are unverified.

### PUBLISHED ROUTE / NO OUTREACH SENT

Official Arrow emergency/trauma inquiry; ask to be routed to the relevant clinical/R&D team.

**P09 / OFFICIAL SOURCE**

**P10 / OFFICIAL SOURCE**

### TAILORED DRAFT PITCH / PREPARED WITH AI / NOT SENT

VASCOUT proposes a passive contact shoe around an ultrasound-led imaging probe, with an open working corridor and proposed force feedback. Your Arrow access portfolio suggests a useful perspective on whether that interface helps or obstructs the clinical workflow. Could this inquiry be routed to a clinical or R&D team willing to review a task-trainer usability protocol? We would measure gloved reach, contact deformation, working-space obstruction, setup burden and cleaning constraints. This is an unvalidated concept; the immediate request is requirements feedback, not a treatment recommendation or a claim of improved access success.

**Official engagement route**

CLINICAL BENEFIT / SAFETY / FIELD SUITABILITY: UNVERIFIED

# Medical Technology Enterprise Consortium

Teaming and funding-route assessment / Proposed research fit is an INFERENCE. Partner interest is unconfirmed.

### SOURCED CAPABILITY / P13

MTEC publishes military-medical teaming opportunities and membership-based funding participation requirements. Its community includes industry, academia and government.

### PROPOSED FIRST STUDY / INFERENCE

Map the concept's validation gaps to an appropriate, currently eligible research opportunity and a team with complementary capability. Review a specific call before preparing any application or assuming award eligibility.

### PROPOSED OUTPUT

A partner-role map and a solicitation-specific eligibility/fit assessment; no funding projection.

### VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

VASCOUT's organizational readiness, membership, fees, call alignment and award eligibility are unverified. No specific open call was audited; membership does not imply funding or validation.

### PUBLISHED ROUTE / NO OUTREACH SENT

Official membership/teaming page; request guidance on fit and prerequisites before enrollment or proposal work.

**P13 / OFFICIAL SOURCE**

### TAILORED DRAFT PITCH / PREPARED WITH AI / NOT SENT

VASCOUT is an unvalidated ultrasound-led vascular-imaging concept with an experimental PA extension. The next need is a credible team and comparative protocol, not a claim of field readiness. Could MTEC advise whether its teaming ecosystem is appropriate for combining an imaging-platform developer, a clinical investigator and a military-medical simulation partner? We would first clarify organizational and membership prerequisites, then assess a specific solicitation's technical scope and eligibility. The proposed research would report incremental benefit and failure modes against ultrasound alone; no funding or award fit is assumed.

**Official engagement route**

CLINICAL BENEFIT / SAFETY / FIELD SUITABILITY: UNVERIFIED

# Sources / clinical and imaging context

Research snapshot: 20 October 2026. URLs link to the retained source record; source evidence does not validate VASCOUT.

S01 / Current DFU / Manufacturer

### Veinlite LEDX - directions for use

Manufacturer DFU. Visible transillumination; not NIR or PA tomography. No comparative efficacy proof.

S02 / 2024 / Healthy human

### Photoacoustic vector tomography for deep haemodynamic imaging

Reported healthy-human vascular imaging. Different research system; full-text access incomplete.

S03 / 2021 / Healthy human

### Photoacoustic imaging of human vasculature using LED versus laser illumination

LED/laser research comparison in phantoms and healthy limbs. No handheld field validation; publisher access limited.

S04 / 2022 / Clinical

### Monitoring peripheral hemodynamic response to changes in blood pressure via photoacoustic imaging

Dialysis/exercise hemodynamics. Not a dehydration meter or shock classifier; root full-text reread incomplete.

S05 / 2022 / Phantom

### Thin ceramic PZT dual- and multi-frequency pMUT arrays for photoacoustic imaging

Sensor-array prior art in agar. No real-time human access proof; direct PMC access challenged.

S06 / 2026 / Healthy human

### The confounding effects of skin colour in photoacoustic imaging

42 healthy participants; optical/acoustic confounding. No demonstrated VASCOUT skin-tone invariance.

S10a / 2005 / Clinical

### Ultrasonography-guided peripheral intravenous access versus traditional approaches

60 difficult-access patients; reported benefit. Odd/even-day allocation, two operators; full text unverified. Read with S10b.

S10b / 2009 / Clinical

### Ultrasonographically guided peripheral intravenous cannulation: a randomized trial

59-patient randomized ED trial; no demonstrated superiority. Workflow-specific; full text unverified. Read with S10a.

# Sources / limitations, engineering and prior art

Research snapshot: 20 October 2026. URLs link to the retained source record; source evidence does not validate VASCOUT.

S07 / 2026 / Guideline

### TCCC Guidelines - vascular access claim audit

1 May 2026 guideline, directly read in the retained audit. Clinical context only; no treatment directions in this deck.

S08 / 2009 / Cadaver

### Bedside ultrasound for detection of soft tissue foreign bodies: a cadaveric study

Cadaver study; small objects and poor reliability. Negative imaging cannot universally exclude fragments.

S09 / 2018 / Simulation

### Ultrasound evaluation of soft-tissue foreign bodies by US Army medics

28 medics, chicken-thigh simulation after training. No injured-human or battlefield validation; retrieval limits retained.

S11 / 2021 / Regulator

### FDA: Imagio Breast Imaging System - P200003

FDA approval for another product and breast-imaging indication. No VASCOUT clearance or pathway follows.

S12 / 2018 / Healthy human

### A device for improving the visual clarity and dimension of veins

20 healthy volunteers in vacuum-device research. Not shock evidence; retrieval gaps. Suction excluded here.

L01 / Manufacturer / Manufacturer

### TI AFE5832LP - ultrasound analog front end

Manufacturer: 32 receive inputs, 16 multiplexed ADC channels. Complete system architecture unresolved.

L02 / Manufacturer / Manufacturer

### Analog Devices MAX2082 - octal ultrasound transceiver

Manufacturer: octal transmit/receive device. Does not prove proposed system integration or power budget.

G01 / 2026 / Concept material

### User-supplied Gemini concepts - audited research inputs

User-supplied Gemini proposals, not studies. Audited concept inputs; no scientific validation. See live evidence library.

# Sources / partner research 1 of 2

Accessed 20 October 2026 EDT / 21 October UTC. Primary organization pages; capability and route evidence only.

P01 **Verasonics - About and photoacoustics capabilities**
Direct official read. Research ultrasound and PA modes; VASCOUT compatibility and benefit unverified.

P02 **Verasonics - Contact and collaborations**
Direct official read. Published collaboration route; no project-specific invitation, resources or acceptance.

P03 **CYBERDYNE - Acoustic X research system**
Direct official read. LED-PA/US research system and vascular-access prior art; novelty cannot be assumed.

P04 **CYBERDYNE - Contact**
Direct official read. Acoustic X inquiry category; no confirmed research partnership or award.

P05 **FUJIFILM Sonosite - Collaborate With Us on Advancing POCUS AI**
Official indexed text only; direct fetch redirect loop. AI-focused, clinician-led scope and current intake need confirmation.

P06 **FUJIFILM Sonosite - Military Medicine Ultrasound**
Official indexed text only; direct fetch redirect loop. Military POCUS context is not VASCOUT field evidence.

P07 **BD - SiteRite 9 Ultrasound System**
Direct official read. Manufacturer vascular-access ultrasound description; no VASCOUT equivalence or clinical-benefit proof.

# Sources / partner research 2 of 2

Accessed 20 October 2026 EDT / 21 October UTC. Primary organization pages; capability and route evidence only.

P08 **BD - Innovation and Idea Submission Process**
Direct official read. Initial idea route is nonconfidential; terms, fit and interest need review. Nothing submitted.

P09 **Teleflex - Emergency/Trauma Products**
Direct official read. Arrow emergency/trauma portfolio; no open imaging collaboration established.

P10 **Teleflex - Request More Information, Emergency/Trauma Products**
Direct official read. General product inquiry only; R&D routing and sponsorship unconfirmed. Form not submitted.

P11 **USAISR - Organ Support and Automation Technologies**
Direct official read. Device development and simulation capabilities; no VASCOUT requirement or endorsement.

P12 **USAISR - Partnering with USAISR**
Direct official read. Public CRADA/technology-transfer route; eligibility, facilities, funding and interest unconfirmed.

P13 **MTEC - Join and membership requirements**
Direct official read. Membership, fees and call-specific eligibility apply; no matched open solicitation audited.

P14 **University of Michigan BME - Xueding Wang and Optical Imaging Laboratory**
Direct official BME profile read. Adjacent imaging methods; lab capacity and collaboration interest unconfirmed. Separate lab homepage returned403.